

3

SPHERICAL SURFACES

Many common optical devices contain not only mirrors and prisms having flat polished surfaces but lenses having spherical surfaces with a wide range of curvatures. Such spherical surfaces, in contrast with plane surfaces treated in the last chapter, are capable of forming real images.

Cross-sectional diagrams of several standard forms of lenses are shown in Fig. 3A. The three *converging*, or *positive*, lenses, which are thicker at the center than at the edges, are shown as (a) *equiconvex*, (b) *plano-convex*, and (c) *positive meniscus*. The three *diverging*, or *negative*, lenses, which are thinner at the center, are (d) *equiconcave*, (e) *plano-concave*, and (f) *negative meniscus*. Such lenses are usually made of optical glass as free as possible from inhomogeneities, but occasionally other transparent materials like quartz, fluorite, rock salt, and plastics are used. Although we shall see that the spherical form for the surfaces may not be the ideal one in a particular instance, it gives reasonably good images and is much the easiest to grind and polish.

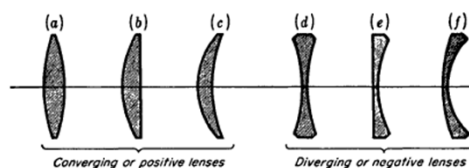


FIGURE 3A
Cross sections of common types of thin lenses.

3.1 FOCAL POINTS AND FOCAL LENGTHS

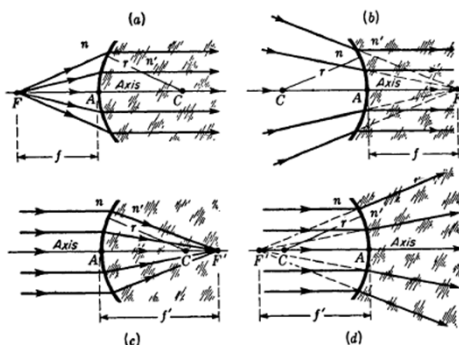


FIGURE 3B
The focal points F and F' and focal lengths f and f' associated with a single spherical refracting surface of radius r separating two media of index n and n' .

It is important to note in Fig. 3B that the primary focal length f for the convex surface [diagram (a)] is not equal to the secondary focal length f' of the same surface [diagram (c)]. It will be shown in Sec. 3.4 that the ratio of the focal lengths f'/f is equal to the ratio n'/n of the corresponding refractive indices [see Eq. (3e)]:

$$\frac{f'}{f} = \frac{n'}{n} \quad (3a)$$

3.2 IMAGE FORMATION

FIGURE 3C

How parallel incident rays are brought to a focus at Q' in the secondary focal plane of a single spherical surface.



A diagram illustrating image formation by a single refracting surface is given in Fig. 3D. It has been drawn for the case in which the first medium is air with an index $n = 1$ and the second medium is glass with an index $n' = 1.60$. The focal lengths f and f' therefore have the ratio 1:1.60 [see Eq. (3a)]. Experimentally it is observed that if the object is moved closer to the primary focal plane, the image will be formed farther to the right away from F' and will be larger, i.e., magnified. If the object is moved to the left, farther away from F , the image will be found closer to F' and will be smaller in size.

3.3 VIRTUAL IMAGES

Since the refracted rays are diverging, they will not come to a focus at any point. To an observer's eye located at the right, however, such rays will appear to be coming from the common point Q' . In other words, Q' is the image point corresponding to the object point Q . Similarly M' is the image point corresponding to the object point M . Since the refracted rays do not come from Q' but only appear to do so, no image can be formed on a screen placed at M' . For this reason such an image is said to be *virtual*.

3.4 CONJUGATE POINTS AND PLANES

The principle of the reversibility of light rays has the consequence that if $Q'M'$ in Fig. 3D were an object, an image would be formed at QM . Hence, if any object is placed at the position previously occupied by its image, it will be imaged at the position previously occupied by the object. The object and image are thus interchangeable, or conjugate. Any pair of object and image points such as M and M' in Fig. 3D are called *conjugate points*, and planes through these points perpendicular to the axis are called *conjugate planes*.



FIGURE 3D
All rays leaving the object point Q and passing through the refracting surface are brought to a focus at the image point Q' .

If one is given the radius of curvature r of a spherical surface separating two media of index n and n' , respectively, as well as the position of an object, there are three general methods that may be employed to determine the position and size of the image: (1) graphical methods, (2) experiment, and (3) calculation using the formula

$$\frac{n}{s} + \frac{n'}{s'} = \frac{n' - n}{r} \quad (3b)$$

In this equation s is the object distance and s' is the image distance. This equation, called the *Gaussian formula for a single spherical surface*, is derived in Sec. 3.10.

3.5 CONVENTION OF SIGNS

The following set of sign conventions will be adhered to throughout the following chapters on geometrical optics, and it would be well to have them firmly in mind:

- 1 All figures are drawn with the light traveling from left to right.
- 2 All object distances (s) are considered positive when they are measured to the left of the vertex and negative when they are measured to the right.
- 3 All image distances (s') are positive when they are measured to the right of the vertex and negative when to the left.
- 4 Both focal lengths are positive for a converging system and negative for a diverging system.
- 5 Object and image dimensions are positive when measured upward from the axis and negative when measured downward.
- 6 All convex surfaces are taken as having a positive radius, and all concave surfaces are taken as having a negative radius.

3.6 GRAPHICAL CONSTRUCTIONS. THE PARALLEL-RAY METHOD

It would be well to point out here that although the above formulas hold for all possible object and image distances, they apply only to images formed by paraxial rays. For such rays the refraction occurs at or very near the vertex of the spherical

The parallel-ray method of construction is illustrated in Figs. 3F and 3G for convex and concave surfaces, respectively. Consider the light emitted from the highest point Q of the object in Fig. 3F. Of the rays emanating from this point in different directions the one (QT) traveling parallel to the axis will by definition of the focal point be refracted to pass through F' . The ray QC passing through the center of curvature is undeviated because it crosses the boundary perpendicular to the surface.

These two rays are sufficient to locate the tip of the image at Q' , and the rest of the image lies in the conjugate plane through this point. All other paraxial rays from Q , refracted by the surface, will also be brought to a focus Q' . As a check we note that the ray QS , which passes through the point F , will (by definition of the primary focal point) be refracted parallel to the axis and will cross the others at Q' as shown in the figure.

This method is called the *parallel-ray method*. The numbers 1, 2, 3, . . . indicate the order in which the lines are customarily drawn.

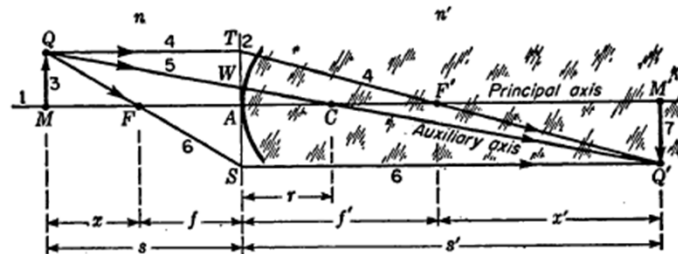


FIGURE 3F
Parallel-ray method for graphically locating the image formed by a single spherical surface.

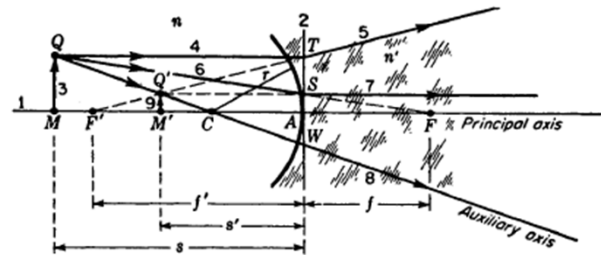


FIGURE 3G
The parallel-ray method applied to a concave spherical surface having diverging properties.

3.7 OBLIQUE-RAY METHODS

Method 1

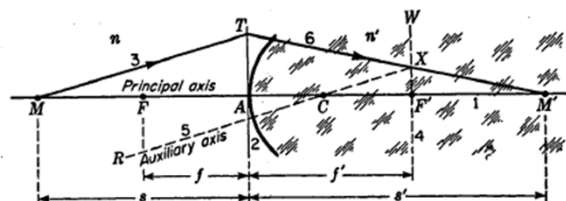


FIGURE 3H
The oblique-ray method for graphically locating images formed by a single spherical surface.

Let MT in Fig. 3H represent any ray incident on the surface from the left. Through the center of curvature C a dashed line RC is drawn, parallel to MT , and extended to the point where it crosses the secondary focal plane. The line TX is then drawn as the refracted ray and extended to the point where it crosses the axis at M' . Since the axis may here be considered as a second ray of light, M represents an axial object point and M' its conjugate image point.

Method 2 This method is shown in Fig. 3I. After drawing the axis MM' and the arc representing the spherical surface with a center C , any line such as 1 is drawn to represent any oblique ray of light. Next, an auxiliary diagram is started by drawing XZ parallel to the axis. With an origin at O , line intervals OK and OL are laid off proportional to n and n' , respectively, and perpendiculars are drawn through K , L , and A . From here the construction proceeds in the order of the numbers 1, 2, 3, 4, 5, and 6. Line 2 is drawn through O parallel to line 1, line 4 is drawn through J parallel to line 3, and line 6 is drawn through T parallel to line 5.

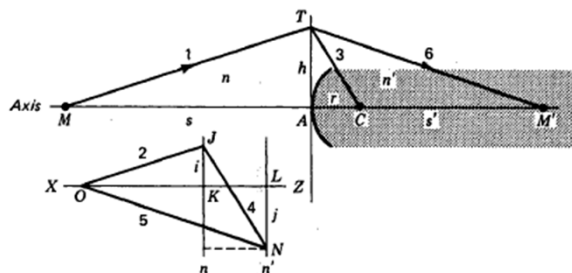


FIGURE 3I
The auxiliary-diagram method for graphically locating images formed by paraxial rays.

3.8 MAGNIFICATION

In any optical system the ratio between the transverse dimension of the final image and the corresponding dimension of the original object is called the *lateral magnification*. To determine the relative size of the image formed by a single spherical surface, reference is made to the geometry of Fig. 3F. Here the undeviated ray 5 forms two similar right triangles QMC and $Q'M'C$.

The theorem of the proportionality of corresponding sides requires that

$$\frac{M'Q'}{MQ} = \frac{CM'}{CM} \quad \text{or} \quad \frac{-y'}{y} = \frac{s' - r}{s + r}$$

We now define y'/y as the lateral magnification m and obtain

$$\bullet \quad m = \frac{y'}{y} = -\frac{s' - r}{s + r} \quad (3g)$$

If m is positive, the image will be virtual and erect, while if it is negative, the image is real and inverted.

3.10 DERIVATION OF THE GAUSSIAN FORMULA

The basic equation (3b) is of sufficient importance to warrant its derivation in some detail. While there are many ways of performing a derivation, a method involving oblique rays will be given here. In Fig. 3K an oblique ray from an axial object point M is shown incident on the surface at an angle ϕ and refracted at an angle ϕ' . The refracted ray crosses the axis at the image point M' . If the incident and refracted rays MT and TM' are paraxial, the angles ϕ and ϕ' will be small enough to permit putting the sines of the two angles equal to the angles themselves; for Snell's law we write

$$\frac{\phi}{\phi'} = \frac{n'}{n} \quad (3k)$$

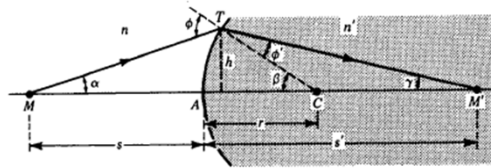


FIGURE 3K
Geometry for the derivation of the paraxial formula used in locating images.

Since ϕ is an exterior angle of the triangle MTC and equals the sum of the opposite interior angles,

$$\phi = \alpha + \beta \quad (3l)$$

Similarly β is an exterior angle of the triangle TCM' , so that $\beta = \phi' + \gamma$ and

$$\phi' = \beta - \gamma \quad (3m)$$

Substituting these values of ϕ and ϕ' in Eq. (3k) and multiplying out, we obtain

$$n'\beta - n'\gamma = n\alpha + n\beta \quad \text{or} \quad n\alpha + n'\gamma = (n' - n)\beta$$

For paraxial rays α , β , and γ are very small angles, and we may set $\alpha = h/s$, $\beta = h/r$, and $\gamma = h/s'$. Substituting these values in the last equation, we obtain

$$n \frac{h}{s} + n' \frac{h}{s'} = (n' - n) \frac{h}{r}$$

By canceling h throughout we obtain the desired equation,

$$\frac{n}{s} + \frac{n'}{s'} = \frac{n' - n}{r} \quad (3n)$$

we have $s' = \infty$, and Eq. (3b) becomes

$$\frac{n}{s} + \frac{n'}{\infty} = \frac{n' - n}{r}$$

Since this particular object distance is called the primary focal length f , we may write

$$\frac{n}{f} = \frac{n' - n}{r} \quad (3c)$$

Similarly, if the object distance is made larger and eventually approaches infinity, the image distance diminishes and becomes equal to f' in the limit, $s = \infty$. Then

$$\frac{n}{\infty} + \frac{n'}{s'} = \frac{n' - n}{r}$$

or, since this value of s' represents the secondary focal length f' ,

$$\frac{n'}{f'} = \frac{n' - n}{r} \quad (3d)$$

Equating the left-hand members of Eqs. (3c) and (3d), we obtain

$$\frac{n}{f} = \frac{n'}{f'} \quad \text{or} \quad \frac{n'}{n} = \frac{f'}{f} \quad (3e)$$

When $(n' - n)/r$ in Eq. (3b) is replaced by n/f or by n'/f' according to Eqs. (3c) and (3d), there results

$$\frac{n}{s} + \frac{n'}{s'} = \frac{n}{f} \quad \text{or} \quad \frac{n}{s} + \frac{n'}{s'} = \frac{n'}{f'} \quad (3f)$$

Both these equations give the conjugate distances for a single spherical surface.

3.9 REDUCED VERGENCE

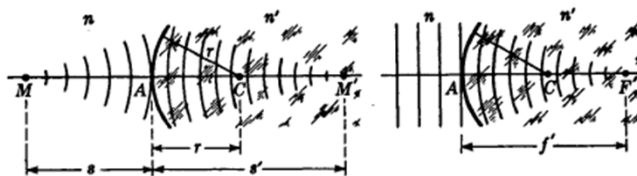


FIGURE 3J
The refraction of light waves at a single spherical surface.

In the formulas for a single spherical refracting surface, Eqs. (3b) to (3f), the distances s , s' , r , f , and f' appear in the denominators. The reciprocals $1/s$, $1/s'$, $1/r$, $1/f$, and $1/f'$ actually represent curvatures of which s , s' , r , f , and f' are the radii.

As the waves from M strike the vertex A , they have a radius s and a curvature $1/s$, and as they leave A , converging toward M' , they have a radius s' and a curvature $1/s'$. Similarly the incident waves arriving at A in the second diagram have an infinite radius ∞ and a curvature of $1/\infty$, or zero. At the vertex where they leave the surface, the radius of the refracted waves is equal to f' and their curvature is equal to $1/f'$.

The gaussian formulas may therefore be considered as involving the addition and subtraction of quantities proportional to the curvatures of spherical surfaces.

When these curvatures rather than radii are used,

$$V = \frac{n}{s} \quad V' = \frac{n'}{s'} \quad K = \frac{1}{r} \quad P = \frac{n}{f} \quad P = \frac{n'}{f'} \quad (3h)$$

The first two, V and V' , are called *reduced vergences* because they are direct measures of the convergence and divergence of the object and image wave fronts, respectively.

For a divergent wave from the object s is positive, and so is the vergence V . For a convergent wave, on the other hand, s is negative, and so is its vergence. For a converging wave front toward the image, V' is positive, and for a diverging wave front, V' is negative. Note that in each case the refractive index involved is that of the medium in which the wave front is located.

The third quantity K is the curvature of the refracting surface (reciprocal of its radius), while the fourth and fifth quantities are, according to Eq. (3e), equal and define the refracting power. When all distances are measured in meters, the reduced vergences V and V' , the curvature K , and the power P are in units called *diopters*. We may think of V as the power of the object wave front just as it touches the refracting surface and V' as the power of the corresponding image wave front which is tangent to the refracting surface. In these new terms, Eq. (3b) becomes

$$V + V' = P \quad (3i)$$

• where

$$P = \frac{n' - n}{r} \quad \text{or} \quad P = (n' - n)K \quad (3j)$$

3.11 NOMOGRAPHY

The term *nomograph* is derived from the greek words *nomos*, meaning *law*, and *graphein*, meaning *to write*. In physics the term applies to certain graphical representations of physical laws, which are intended to simplify or speed up calculations. Figure 3L is a nomograph relating *object* and *image distances* as given by Eq. (3f), namely,

$$\frac{n}{s} + \frac{n'}{s'} = \frac{n}{f} \quad (3o)$$

Its simplicity and usefulness become apparent when it is seen that any straight line drawn across the figure will intersect the three lines at values related by the above equation.

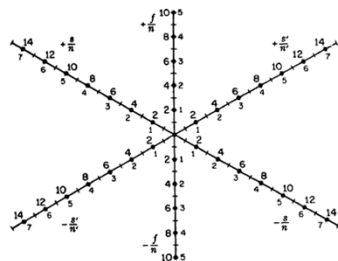


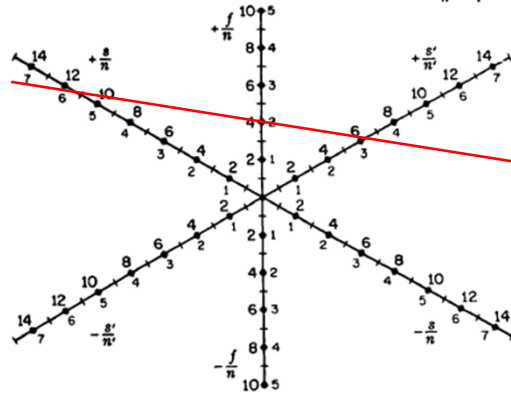
FIGURE 3L
Nomograph for determining the object or image distance for a single spherical surface or for a thin lens.

A little study of this nomograph will show that it applies to all object and image distances, real or virtual, and to all surfaces with positive or negative radii of curvature.

EXAMPLE 4 One end of a plastic rod of index 1.5 is ground and polished to a radius of +2.0 cm. If an object in air is located on the axis 12.0 cm from the vertex, what is the image distance?

SOLUTION The given quantities are $n = 1.0$, $n' = 1.50$, $r = +2.0$ cm, and $s = +12.0$ cm. The unknown is s' . By direct substitution and the use of Eq. (3o) we obtain

$$\frac{s}{n} = \frac{12}{1} = +12.0 \quad \text{and} \quad \frac{f}{n} = \frac{r}{n' - n} = \frac{2}{1.5 - 1} = +4.0$$



If the straight edge of a ruler is now placed on $s/n = +12.0$ and $f/n = +4.0$, it will intersect the third line at $s'/n' = +6.0$. Since $n' = 1.5$, s' is equal to 6×1.5 , or +9.0 cm.